

STA304H5F: Surveys, Sampling and Observational Data

1. Project Title:

Understanding the Impact of Academic Workload on Student Mental Health (38)

2. Research questions (2-4)

1. Is there a relationship between students' academic workload and amount of mental health issues?
2. How does the academic workload affect the quality/length of sleep and its subsequent impact on students' mental health?
3. Is mental health affected by students' living situations or students' ability to maintain work-life balance?

3. Precisely define your target population. (If your study involves collecting data from human participants, the population should be restricted exclusively to STA304H5 students.)

The target population for all three research questions consists of the students enrolled in the STA304H5 course during the Fall 2024 semester.

4. Name the sampling method you will use (SRS, STRS, Cluster, etc). If necessary, describe clusters and/or strata you will use.

Simple Random Sampling. Both sections are merged, and a sample of n is chosen randomly from the combined pool of students.

5. Describe how you plan to collect your data.

Survey in Google Form link and distributed by TA on Piazza.

6. Mention which statistical tests you will use and their purpose in your project. You must pick at least one "simple" type and one "advanced" type.

Simple Type:

1. Chi-square Test for Independence: Study the association between categorical variables:
 - a. Sleep and Living situation
 - b. Stress symptoms and living situation

Advanced Type:

1. Multiple linear regression: Determine the combined effects of academic workload, sleep quality/length, and work-life balance on mental health (dependent variable).

STA304H5F: Surveys, Sampling and Observational Data

2. ANOVA: Compare the average levels of stress in categories of academic workload, sleep quality and work-life balance.

7. **Attach a copy of your questionnaire. One of the questions must ask for course specific IDs for students to obtain marks for completing questionnaires. You may add an optional question for students' emails or other information if you plan to give an incentive. Not counting the previous two questions, the questionnaire should contain 8-10 questions.**